1)

$z_m = \theta_1 x + \theta_2 y + \theta_3 xy$

$w_m = \theta_4 x + \theta_5 y + \theta_6 y^2$

$e = z - z_m = \sum e^2 = (z - \theta_1 x - \theta_2 y - \theta_3 xy)^2$

$\frac{dS}{d\theta_1} = 2 \sum (z - \theta_1 x - \theta_2 y - \theta_3 xy)(-x) \Rightarrow 0 = -\sum xz + \sum \theta_1 x^2 + \sum \theta_2 xy + \sum \theta_3 x^2 y$

$\frac{dS}{d\theta_2} = 2 \sum (z - \theta_1 x - \theta_2 y - \theta_3 xy)(-y) \Rightarrow 0 = -\sum yz + \sum \theta_1 xy + \sum \theta_2 y^2 + \sum \theta_3 xy^2$

$\frac{dS}{d\theta_3} = 2 \sum (z - \theta_1 x - \theta_2 y - \theta_3 xy)(-xy) \Rightarrow 0 = -\sum xyz + \sum \theta_1 x^2 y + \sum \theta_2 xy^2 + \sum \theta_3 x^2 y^2$

$$\begin{pmatrix} \sum x^2 & \sum xy & \sum x^2 y \\ \sum xy & \sum y^2 & \sum xy^2 \\ \sum x^2 y & \sum xy^2 & \sum x^2 y^2 \end{pmatrix} \cdot \begin{pmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{pmatrix} = \begin{pmatrix} \sum xz \\ \sum yz \\ \sum xyz \end{pmatrix}$$

$\sum e^2 = \sum w - \theta_4 x - \theta_5 y - \theta_6 y^2$

$$\begin{pmatrix} \sum x^2 & \sum xy & \sum xy^2 \\ \sum xy & \sum y^2 & \sum y^3 \\ \sum xy^2 & \sum y^3 & \sum y^4 \end{pmatrix} \cdot \begin{pmatrix} \theta_4 \\ \theta_5 \\ \theta_6 \end{pmatrix} = \begin{pmatrix} \sum wx \\ \sum wy \\ \sum wy^2 \end{pmatrix}$$

$\theta_1 = 3 \quad \theta_4 = -3$

$\theta_2 = -4 \quad \theta_5 = 4$

$\theta_3 = 2 \quad \theta_6 = 1$

$3x - 4y + 2xy = 10$

$-3x + 4y + 1y^2 = -8,75$

$3x - 4x + 2x^2 - 10 = 0$

$-3x + 4x + x^2 - 8,75 = 0$

$2x^2 - x - 10 = 0$

$x^2 + x - 8,75 = 0$

$x = y = 2,5$

2)

$V = R \cdot I$

$V = \theta \cdot I$

$I = \frac{1}{R} \cdot V + \theta_2$ (ruido de medição)

$S = \sum (I - \frac{1}{R} V)^2$

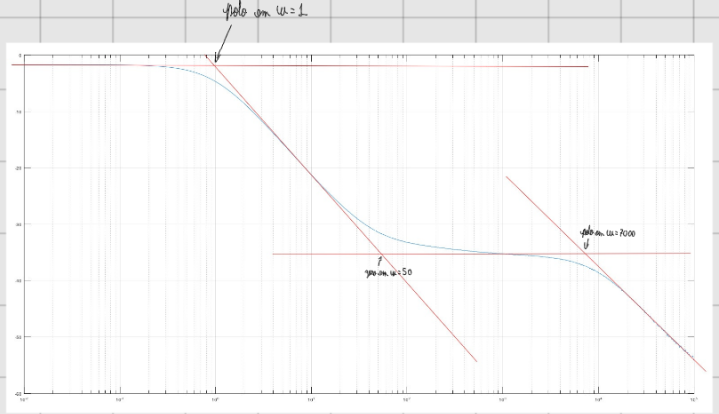
$S \sum -V 2 (I - \theta V)$

$\sum = 2 \sum Vi - \theta V^2$

$\sum Vi - \sum \theta V^2 = 0$

$\sum Vi = \theta \sum V^2$

$\theta = \frac{\sum V \cdot I}{\sum V^2}$



$H(s) = \frac{A \cdot (s+50)}{(s+1)(s+7000)}$ $s/s = 0$

$0.8 = \frac{A \cdot (50)}{1 \cdot 70000} \Rightarrow A = \frac{0.8 \cdot 7000}{50} = 112$ $H(s) = \frac{112 \cdot (s+50)}{(s+1)(s+7000)}$

$\begin{bmatrix} \beta_{V_2} & \beta_{U_2} \\ \beta_{U_1 V} & \beta_{U_1 U_2} \end{bmatrix} \cdot \begin{bmatrix} 1 \\ U_1 \\ U_2 \end{bmatrix} = \begin{bmatrix} -7 - \beta_{V_1} \\ 4 - \beta_{V_2} \end{bmatrix}$

B U Y

$B \cdot U = Y$

$U = B^{-1} \cdot Y$

$H(s) = \frac{Y(s)}{X(s)}$ $X(s) = 2 \cdot w(s)$ $X(s) = \frac{2}{s}$

$H(s) = Y(s) \cdot X(s) = \frac{A(s+B)}{(s+1)} \cdot \frac{2}{s} = \frac{2A(s+B)}{s(s+1)}$

$\frac{2A(s+B)}{s(s+1)} = \frac{a}{s+1} + \frac{b}{s} \cdot s(s+1)$

$2As + 2AB = as + b(s+1)$

$s = 0$

$2Ab = b$

$s = -1$

$-2A + 2AB = -a$

$a = 2A - 2AB \Rightarrow 2A(1-B)$

$Y(s) = \frac{2A(1-B)}{s+1} + \frac{2AB}{s}$

$Y(s) = \underbrace{2A(1-B)}_{\theta_1} \cdot e^{-t} + \underbrace{2AB}_{\theta_2}$

$2 \cdot AB = \theta_2$

$2A - 2AB = \theta_1$

$\begin{bmatrix} 0 & 2 \\ 2 & -2 \end{bmatrix} \cdot \begin{bmatrix} A \\ AB \end{bmatrix} = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}$

$\begin{bmatrix} A \\ AB \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 2 & -2 \end{bmatrix}^{-1} \cdot \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}$

$$\frac{k(s+a)}{(s+b)(s+c)^2}$$

+3 Db em a

-3 Db em b

-6 Db em c

em $s=0 \Rightarrow H(0) = 9.54\,DB = 3$

12.54 Db na freq. 0.01 rad/s

$a = 0.01$

pela assintota

$b = 1$

ganho cont em $10^2 \Rightarrow 49.4$ (43.4)

ganho de -6 Db em $c = 100$

$$H(0) = 3 = k \cdot \frac{(0.01)}{1 \cdot 100^2}$$

$$\frac{100^2 \cdot 3}{0.01} = k$$

$k = 3$